Unit 1: Database Design

Normal forms reduce redundancy in relational databases. A relation is in third normal form when every non-key attribute depends only on the key.

Unit 2: Database Design

Decomposition must keep dependencies and remain lossless so that the original relation can be rebuilt from its parts.

The regula-
tion applies to all labs.

Course	Credits
DBMS	4
Operating Systems	3

Unit 3: Database Design

Transactions group operations into one unit of work. Isolation levels decide which anomalies a concurrent transaction can observe.

Unit 4: Database Design

Indexes speed up lookups at the cost of slower writes. A B-tree keeps keys sorted, so it serves range queries as well as point queries.